Paper Framework Overview

Breast cancer remodels lymphatics in sentinel lymph nodes (Nat Commun, 2025)

1. Clinical question

How does breast cancer metastasis
remodel lymphatic endothelial cells (LECs)
in human sentinel lymph nodes?

2. Human paired samples

9 treatment-naive breast cancer patients
paired distant LN vs metastatic LN
FACS-sorted PDPN+ CD31+ LECs

3. Core profiling

scRNA-seq
Visium HD spatial transcriptomics
immunostaining

4. Computational analysis

Seurat integration
LEC clustering
DEG analysis

5. State transition analysis

miloR differential abundance
Monocle trajectory inference
subset shift in metastatic LN

6. Mechanism prediction

NicheNet sender-receiver modeling
TGF-beta / VEGF-A / VEGF-C
as top upstream ligands

7. Main discoveries

Metastatic LN loses PD-L1+ SCS floor LECs
and medullary sinus LECs,
while gaining capillary-like
CD200+ HEY1+ LECs.

8. Validation in vitro

Breast cancer conditioned media
reprogram LN-derived HLECs.
MGP is consistently induced
across multiple models.

9. Functional model

TGF-beta / VEGF -> MGP upregulation on LECs
MGP increases cancer cell adhesion
to lymphatic sinuses and may support
systemic dissemination.

Take-home message: lymphatics in metastatic sentinel nodes are actively reprogrammed,
not passive conduits, and MGP emerges as a candidate mediator of tumor spread.